

Sonic Needle: Volunteer Guide

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About the demo

Visitors steer a giant ultrasonic needle through a draped box (the patient) to a hidden target, using only the echoes shown on screen. When the needle touches the target, an LED lights, a buzzer sounds and the screen celebrates.

The needle is supposed to be a giant version of one that might be used to take a biopsy of a cancer, for example in the liver or breast.

How it works

- The needle contains an ultrasound transmitter and receiver (a US-100 sensor, 40 kHz, too high-pitched to hear).
- About 20 times a second, the sensor sends a pulse of sound. The pulse bounces off the target and the echo comes back.
- An Arduino detects the echo and sends the arrival time to the laptop over USB. The laptop works out the distance from the speed of sound (343 m/s in air).
- The calculated distance to the target is visualized.
- The box is lined with foam that absorbs sound, so the only echo comes from the target.
- Touching the target closes a circuit. This lights the LED, sounds the buzzer and tells the Arduino the target has been reached.

How to use it (once its running)

- Pull the needle upwards as high as it will go (**it can't be fully removed from the box**).
- Slowly lower the needle into the box while changing the direction in which it is pointing, until the software detects echoes. **THIS WILL AUTOMATICALLY START THE TIMER***
- Move the needle down in the direction of the echoes, readjusting the direction if you lose signal.
- When you reach the target, a buzzer will sound and you'll be shown how long it took you to reach it.
- If you follow the echoes to reach the bottom without the buzzer sounding, try moving side to side slightly – maybe you just missed it.

**you can restart the timer manually by pressing the R key*

Starting the demo

1. Check the Arduino's USB cable is plugged into the laptop. The Arduino's power light should be on.
2. Double-click **Sonic Needle** on the desktop. After a few seconds the display fills the screen.

3. Check the label at the top right says **Sensor live**, with a green dot. If it says **Demo mode: simulated sensor**, you've opened the practice version: press Cmd+Q and open **Sonic Needle** instead.
4. Test it: push the needle into the box. The echo trace should move and the distance should drop.
5. Touch the needle on the target. The LED and buzzer should go off and the screen should show **Target reached!**
6. Check the laptop volume is up. If **Click anywhere to turn on sound** appears at the bottom, click once.

Best times and target count

Daily and all-time records for the time taken to reach the target, as well as the number of times the target has been reached, are shown on screen. You can reset these by pressing Shift-R (today's records and counts) and Shift-A (all-time records and counts).

To shut down, press **Cmd+Q**. This closes the display and stops the software.

Keyboard shortcuts

Key	What it does
R	Reset the current round, e.g. if a visitor walks away
M	Mute or unmute the sound
F	Leave or return to full screen
Shift+R	Clear today's best time and target count
Shift+A	Clear all records.
D	Show Arduino connection details
Cmd+Q	Quit the demo

If it stops working

Most problems are fixed by reconnecting the Arduino or restarting the app; work through these in order.

If the Arduino comes unplugged

The label at the top right turns red and says **No Arduino found - check the USB cable**.

1. Plug the USB cable back in, into any USB port on the laptop.
2. Wait about 5 seconds. The label should return to **Sensor live**.

You don't need to restart the app: it finds the Arduino again by itself.

If the display freezes, stays red or behaves oddly

1. Press **Cmd+Q** to quit.
2. Wait 5 seconds, then double-click **Sonic Needle** again.
3. Still not working? Quit, unplug the Arduino's USB cable, wait 5 seconds, plug it back in, then open **Sonic Needle** again.

If none of this works, check the wiring and the symptoms below, then contact the research team.

Troubleshooting

What you see	Likely cause	What to do
Red label: No Arduino found - check the USB cable	USB cable unplugged or loose	Plug it back in and wait 5 seconds. Try another USB port
Red label: Display not connected	The software has stopped	Press Cmd+Q and open Sonic Needle again
Red label mentioning NewScientistLive.ino	Arduino not running the demo's program	Contact the Christian Baker (or if you're comfortable using an Arduino, then open the .ino file in ~/Documents/SonicNeedle in the Arduino IDE app and send the program to the Arduino).
Sensor live , but the screen shows NO ECHO	Needle tilted, so the echo misses the sensor	Hold the needle upright. Check the target hasn't moved
Buzzer and LED work, but no Target reached! on screen	Blue wire loose	Push the loose blue wire into pin 2
Target reached! on screen, but no buzzer or LED	Red wire loose	Push the loose red wire into 3.3V
Neither works when the needle touches the target	Both case wires loose, or contact not made	Check both case wires. Make sure the tip really touches the target
Target reached! without touching the target	Blue wire loose or touching another pin	Reseat the blue wire in pin 2, away from other pins
No sound from the laptop	Volume down, muted, or sound not yet allowed	Turn the volume up, press M, or click the screen once

Arduino wiring

The needle's four sensor wires are soldered to the Arduino shield and shouldn't come loose. The two wires from the case are only pushed into the shield's connectors, so check them first.

Wire	Comes from	Shield connector	Fixed how	Job
Red	Case	3.3V	Pushed in	Powers the buzzer and LED
Blue	Case	Pin 2	Pushed in	Tells the Arduino the needle has touched the target
Red, green tape, soldered	Needle	5V	Soldered	Powers the ultrasound sensor
Blue, soldered	Needle	GND	Soldered	Ground for the sensor
Yellow	Needle	Pin 11	Soldered	Tells the sensor to send a pulse
Orange	Needle	Pin 12	Soldered	Carries the echo timing back

Talking about the research

The demo shows how ultrasound small enough to fit inside a needle could make smart needles that track, navigate and image tissue.

- **The problem.** Needles are used for biopsies, injections, nerve blocks and draining fluid. Doctors often guide them with an ultrasound scanner, but a thin needle tip can be hard to see on the image.
- **Our approach.** We make ultrasound transmitters and sensors on optical fibres a fraction of a millimetre across, small enough to fit inside a needle or other surgical instrument.
- **Tracking.** A sensor at the tip can pick up pulses from an ultrasound scanner, so the tip's position can be shown on the ultrasound image. Or, the needle tip can transmit ultrasound to an external array, which can calculate its location.
- **Imaging.** With a transmitter and a sensor together, or using a fibre to transmit light into the tissue, generating sound via the photoacoustic effect, the needle can make images of the tissue right in front of its tip.
- **Demo versus the real thing.** The demo's sensor is a couple of centimetres across and works in air. Ours are hundreds of times smaller, use much higher frequencies and work inside the body.